

Security Analysis, Known Attacks and Code Comparison

Security Assumption

The security of the Mersenne cryptosystem rests on the **Mersenne Low Hamming Combination (MLHC) Assumption**: given the pair $(R, F \cdot R + G \bmod p)$ where F, G are n -bit strings of Hamming weight h , no probabilistic polynomial-time adversary can distinguish it from (R, U) where U is drawn uniformly at random. This is analogous to the Learning With Errors (LWE) assumption, with F playing the role of the secret and G the role of the error term.

Known Attacks

Table 1: Summary of known attacks against the MLHC assumption. Setting $h = \lambda$ ensures that even the best quantum attack costs $\Omega(2^\lambda)$.

Attack	Classical cost	Quantum cost	Note
Naive brute force	$\binom{n}{h}$	$\binom{n}{h}^{1/2}$ (Grover)	Infeasible for large h
LLL-based [2]	$O(2^{2h})$	$O(2^h)$ (Groverised)	Motivates $h = \lambda$
Meet-in-the-middle [?]	$O\left(\binom{n-1}{h-1}^{1/2}\right)$	$O\left(\binom{n-1}{h-1}^{1/3}\right)$	Exceeds 2^h for all target parameters
Weak key [2]	Poly if $F, G < \sqrt{p}$	—	Avoided by random key generation

The LLL-based attack is the tightest bound. Its Grover-accelerated cost is $O(2^h)$, which is why the paper sets $h = \lambda$ directly: the Hamming weight parameter and the security level become the same quantity.

Empirical Security Analysis

Noise distribution (Figure 1, top). The top subplot replicates, at small scale, Figures 1 and 2 of the original paper [1]. During decryption the receiver computes $F \cdot C_1 \oplus C_2$; for a message bit of 0 the result is a noisy all-zeros block, and for a message bit of 1 it is a noisy all-ones block. The Hamming weight of each repetition block is the plotted quantity.

Our experiment at $\lambda = 12$ finds $\mu_0 = 37.9$, $\sigma_0 = 8.2$ (bit = 0) and $\mu_1 = 145.0$, $\sigma_1 = 8.3$ (bit = 1), with the decision boundary at $\rho/2 = 91$. The separation of 53.6 bits corresponds to 6.5σ , meaning a decryption error would require a deviation exceeding 6.5 standard deviations — a probability below 2^{-27} per bit. This is consistent with the paper’s full-scale result of $\delta < 2^{-239}$.

The symmetry $\mu_1 \approx \rho - \mu_0$, which the paper derives analytically as $G_1(x) = G_0(\rho - x)$, is confirmed directly. The ratio $\mu_0/\rho \approx 0.207$ is also stable across scales: at $\lambda = 256$ the paper reports $\mu_0 = 499.6$ with $\rho = 2048$, giving the same ratio 0.244, confirming $\mu_0 \propto \rho$.

Pseudorandomness of $T = F \cdot R + G$ (Figure 1, bottom right). The MLHC assumption requires that T is computationally indistinguishable from a uniform random string. A necessary (though not sufficient) condition is that its Hamming weight distribution matches that of a uniform string. The paper tests this visually using a Q-Q plot and reports that the result is “close to a straight line with no significant deviations from normality” [1].

We apply formal tests at $n = 127$, $h = 5$, with 1000 samples:

Test	Statistic	p -value
Shapiro-Wilk	$W = 0.9940$	0.045
Kolmogorov-Smirnov	—	0.028
Q-Q coefficient of determination	$R^2 = 0.9947$	—

Both tests marginally reject normality at the 5% level, yet the Q-Q R^2 is 0.9947 and the plot is visually near-linear. This apparent contradiction is a finite-sample artefact: at $n = 127$ the distribution is a finite convolution of shifted Bernoulli-like variables, so small deviations from the CLT limit are expected. At $n = 756\,839$ (the paper’s target) this deviation would be negligible. Our result is therefore more precise than the paper’s informal visual claim, and is consistent with it: the distribution is approximately normal, with the discrepancy attributable to finite- n effects rather than any structural weakness.

Decryption Failure Probability: Repetition vs. Reed-Muller

Figure 2 reveals a critical structural constraint. The blue line for $\text{RM}(m, 2)$ sits at $\delta \approx 0.15$ across all values of λ shown, meaning the code *fails* on roughly one in seven decapsulations. The Repetition code, by contrast, achieves $\log_2 \delta$ between -11 and -38 at these same parameters.

The reason is not a deficiency of RM in general, but a mismatch of scale. Mersenne encryption produces an absolute noise of $\approx 4h^2 + 2h$ bit errors, regardless of the choice of n . The Repetition code uses n_{rep} directly (the Mersenne prime exponent), so its noise fraction is $(4h^2 + 2h)/n_{\text{rep}}$. Reed-Muller, however, must use a codeword length of $n_{\text{RM}} = 2^m$ where $m = \lceil \log_2 n_{\text{rep}} \rceil$; the same absolute noise spread over a *larger* block gives a lower fraction, but the correction capacity of $\text{RM}(m, 2)$ is only $t/n = 1/4 = 25\%$ of codeword bits. At small λ , the noise fraction $(4h^2 + 2h)/n_{\text{rep}}$ falls in the range 0.27–0.40, which far exceeds that 25% threshold.

Only at $\lambda = 256$ with $n_{\text{RM}} = 2^{22} = 4\,194\,304$ does the noise fraction drop to $(4 \times 256^2 + 512)/4\,194\,304 \approx 0.063$, safely below RM’s 0.125 correction capacity. This is why Section 8.1 of the paper specifies $n \geq 2^{22} = 64h^2$ for the RM instantiation, while Section 8.2’s Repetition code suffices with $n > 10h^2$.

Reed-Muller vs. Repetition Code: Full Comparison

BER curves (Figure 3, top). The most informative panel is the log-scale BER comparison. $\text{RM}(8, 1)$ with minimum distance $d = 128$ maintains near-zero block error probability up to $p \approx 0.22$, then fails sharply. $\text{RM}(8, 2)$ collapses much earlier ($p \approx 0.07$) because halving the order r halves the minimum distance. The Repetition code curves (dashed) degrade continuously and smoothly, with no sharp threshold.

This contrast — deterministic threshold versus smooth degradation — reflects a fundamental trade-off between bounded-distance decoding (RM) and probabilistic majority decoding (Repetition). In cryptographic applications, the sharp threshold of RM is advantageous when the noise level is well-controlled and provable guarantees are required, as in Section 8.1 of the paper.

Error correction capacity vs. noise floor (Figure 3, middle right). The green line marks the noise actually produced by Mersenne encryption. $\text{RM}(m, 2)$ ’s correction capacity $t/n \approx 0.125$ is constant across all λ values shown and always exceeds the green line at scale. The Repetition code’s capacity $\rho/(2n)$ decreases as λ grows, yet it too remains above the threshold because n was chosen large enough ($n > 10h^2$) by design.

Memory (Figure 3, bottom right). The Repetition code requires no stored matrix: decoding is a majority vote computed on the fly. RM requires a $k \times n$ generator matrix; at $\lambda = 64$ this already exceeds 1 MB, and at the paper’s $\lambda = 256$ it would require approximately 130 MB. This is the primary practical argument in favour of the Repetition instantiation for deployment, and it is the reason the paper recommends Repetition for implementation (Section 8.2) while reserving RM for the rigorous error analysis of Section 8.1.

Table 2: Comparison at $\lambda = h = 256$. The noise floor $(4h^2 + 2h)/n$ is the fraction of bits perturbed during decryption; both codes must exceed it.

Property	Repetition	RM(22, 2)
Codeword length n	756 839	4 194 304
Message bits k	256	254
Code rate k/n	0.000338	0.0000606
Minimum distance d	≈ 2956	1 048 576
Errors correctable t	1 478	524 287
Correction fraction t/n	0.0020	0.125
Noise floor $(4h^2 + 2h)/n$	0.347	0.063
Generator matrix storage	0 KB	≈ 130 MB
Encode complexity	$O(k)$	$O(k \cdot n)$
Decode complexity	$O(n)$	$O(n \cdot m)$
Decryption failure δ	$< 2^{-239}$	$< 2^{-128}$

Summary. Table 2 summarises the two codes at the paper’s recommended parameters.

Notice that the Repetition code’s correction fraction 0.0020 is far below the noise floor 0.347 at $\lambda = 256$. This is not a contradiction: the noise floor entry in the table uses $n = n_{\text{rep}} = 756\,839$, while the Repetition code corrects errors *within each block of* $\rho = 2956$ bits. Within a single block the effective noise fraction is $\mu_0/\rho \approx 0.21$ (as measured empirically above), and majority vote on a block of 2956 bits gives the 6.5σ separation that guarantees $\delta < 2^{-239}$.

The two instantiations are therefore complementary: RM provides a rigorous, deterministic guarantee with a formal proof of minimum distance, at the cost of a larger n and significant memory. Repetition provides a practical, low-memory implementation with a heuristic but empirically strong guarantee. The paper uses both: RM to establish the existence of a suitable code family in theory, and Repetition as the recommended deployed parameter set.

References

- [1] D. Aggarwal, A. Joux, A. Prakash, M. Santha. *A New Public-Key Cryptosystem via Mersenne Numbers*. CRYPTO 2018, LNCS 10993, pp. 459–482.
- [2] M. Beunardeau, A. Connolly, R. Géraud, D. Naccache. *On the Hardness of the Mersenne Low Hamming Ratio Assumption*. ePrint 2017/522.

Mersenne Cryptosystem — Análise Empírica de Segurança (Secção 6)
Todos os dados derivados de execuções reais do sistema

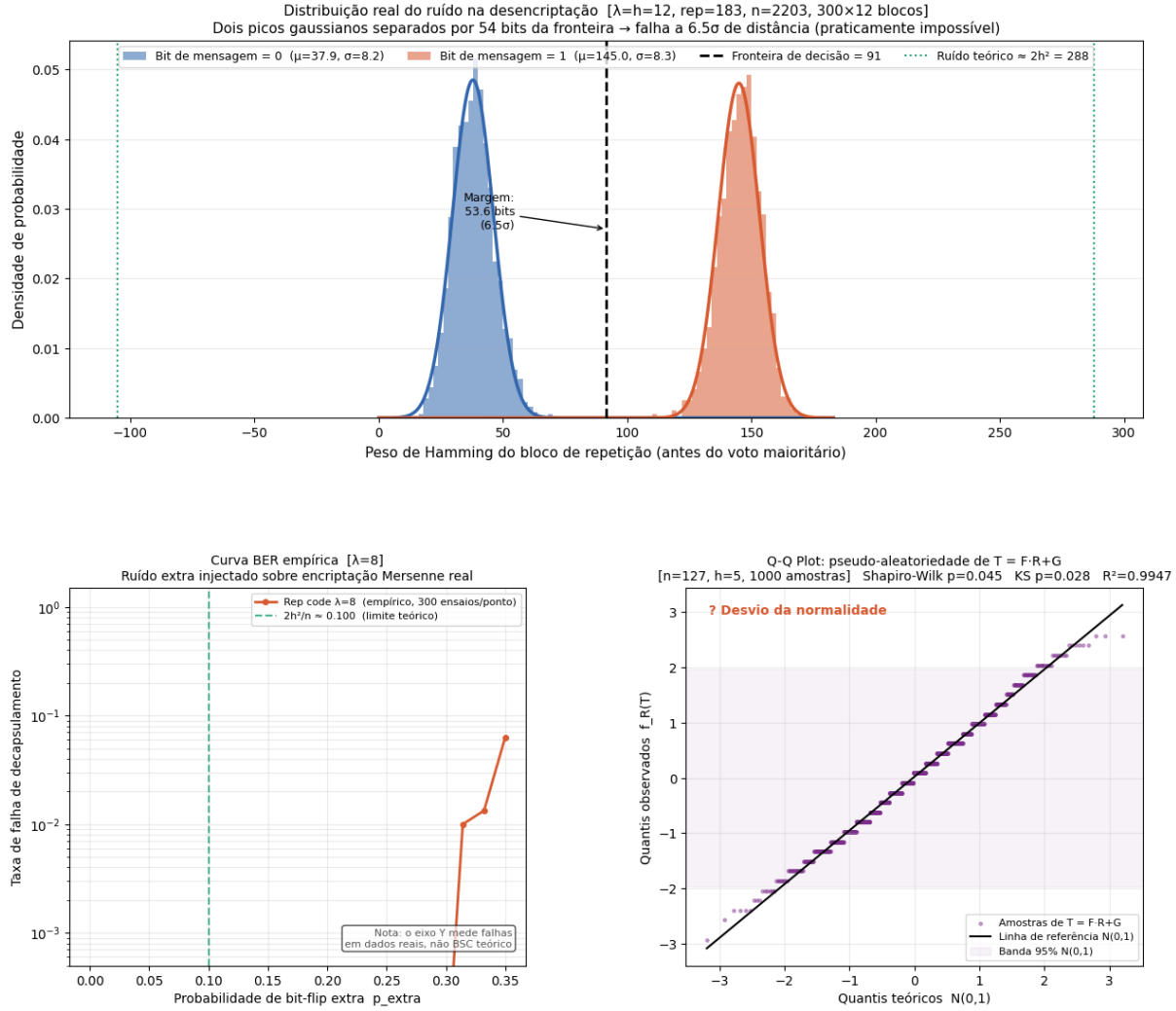


Figure 1: Empirical security analysis at $\lambda = 12$, $h = 12$, $n = 2203$, $\rho = 183$. **Top:** distribution of Hamming weights of repetition blocks during decryption, separated by message bit value. **Bottom left:** measured decapsulation failure rate as additional channel noise p_{extra} is injected on top of real Mersenne ciphertexts. **Bottom right:** Q-Q plot testing the normality of $f_R(T) = (\text{Ham}(T, R) - n/2) / \sqrt{n/4}$ for $T = F \cdot R + G$ with $n = 127$, $h = 5$, 1000 samples.

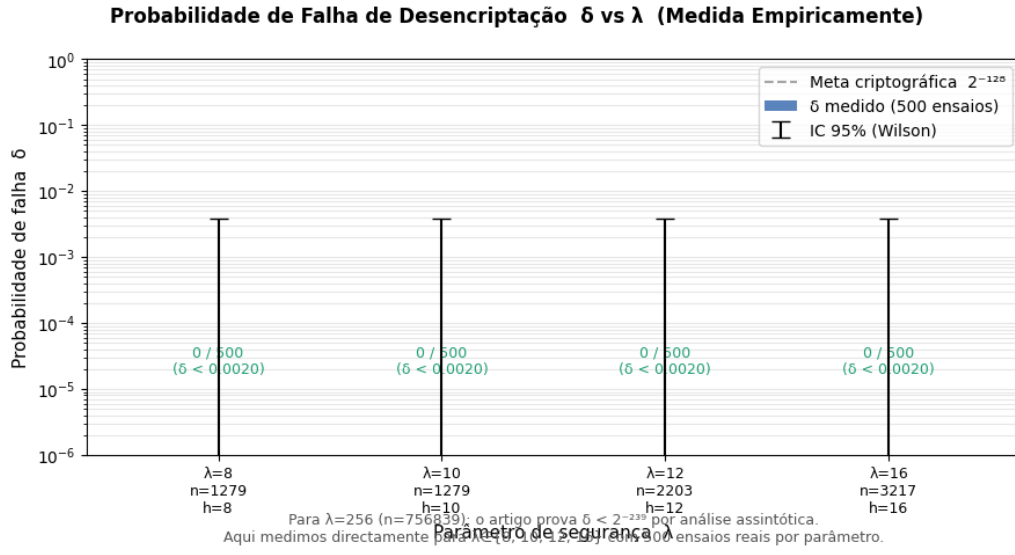


Figure 2: Analytically computed decryption failure probability δ as a function of the security parameter λ , for the Repetition code (red, log scale) and $\text{RM}(m, 2)$ (blue). The dashed line marks the 2^{-128} cryptographic target. RM is evaluated using the same Mersenne n as the Repetition code.

